



04.21 M: How to Train Your Network (Backpropagation)

Completion requirements

- Topics: **How to Train Your Network**
 - Backpropagation (parameters = weight + bias for every unit)
 - Hyperparameters include learning rate, dropout rate, and batch size
 - Network architecture includes weight init, activation function, loss function, number of hidden layers, number of units for each layer, optimizer, dropout layer
- Reading:
 - [Guide to Hyperparameters Search for Deep Learning Models](#)
- Resources:
 - [How to Choose Loss Functions When Training Deep Learning Neural Nets](#) (Machine Learning Mastery)
- **Video Series on Deep Learning (3Blue1Brown)**
 - *These are the **best** videos that I have come across on this topic. The first video in the series mostly recaps ideas that you already know from the content we covered in the last class, but it is still worth watching because it provides such a nice overview. The next two videos explain how we train a neural network. They are highly accessible even if you haven't had linear algebra or calculus. The last video is optional, but encouraged for DSC majors and for anyone who wants to go deeper. It gets into concepts such as gradients and requires a deeper knowledge of calculus. When we meet this afternoon, I will summarize some of this content, but mostly talk about the practical aspects of training a network.*
 - **Deep Learning 1/4: But what is a Neural Network?** (mostly review) [Deep Learning 1/4: But what is a Neural Network?](#)
 - **Deep Learning 2/4: Gradient Descent, How Neural Networks Learn** [Deep Learning 2/4: Gradient Descent, How Neural Networks Learn](#)
 - **Deep Learning 3/4: What Is Backpropagation Really Doing?** [Deep Learning 3/4: What is backpropagation really doing?](#)
 - **Deep Learning 4/4: Backpropagation calculus** (optional, more technical) [Deep Learning 4/4: Backpropagation calculus](#)

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